



A Genomic Analysis Workflow for Colorectal Cancer Precision Oncology

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Abstract

Accurate diagnosis and precision medicine of colorectal cancer (CRC) rely on patient-specific genomic maps. We present IDEA, an integrated DNA next generation sequencing and bioinformatic approach to determine the molecular landscape of CRC. First, genomic targets are predefined to obtain optimal sensitivity for tissue or blood samples. IDEA then pinpoints genetic variations with predictive and prognostic value, defines actionable targets, and unveils drug resistance mechanisms in patients with metastatic CRC. Results are presented in a final report, which includes clinically relevant information.

Background: The diagnosis of colorectal cancer (CRC) is routinely accomplished through histopathologic examination. Prognostic information and treatment decisions are mainly determined by TNM classification, first defined in 1968. In the last decade, patient-specific CRC genomic landscapes were shown to provide important prognostic and predictive information. Therefore, there is a need for developing next generation sequencing (NGS) and bioinformatic workflows that can be routinely used for the assessment of prognostic and predictive biomarkers. **Materials and Methods:** To foster the application of genomics in the clinical management of CRCs, the IDEA workflow has been built to easily adapt to the availability of patient specimens and the clinical question that is being asked. Initially, IDEA deploys ad-hoc NGS assays to interrogate predefined genomic target sequences (from 600 kb to 30 Mb) with optimal detection sensitivity. Next, sequencing data are processed through an integrated bioinformatic pipeline to assess single nucleotide variants, insertions and deletions, gene copy-number alterations, and chromosomal rearrangements. The overall results are gathered into a user-friendly report. **Results:** We provide evidence that IDEA is capable of identifying clinically relevant molecular alterations. When optimized to analyze circulating tumor DNA, IDEA can be used to monitor response and relapse in the blood of patients with metastatic CRC receiving targeted agents. IDEA detected primary and secondary resistance mechanisms to *ERBB2* blockade including sub-clonal *RAS* and *BRAF* mutations. **Conclusions:** The IDEA workflow provides a flexible platform to integrate NGS and bioinformatic tools for refined diagnosis and management of patients with advanced CRC.

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Introduction

Colorectal cancer (CRC) is the third most frequently diagnosed and the second most common cause of cancer death worldwide.¹ It arises from the sequential transformation of the normal intestinal epithelium into benign adenoma and then into an invasive adenocarcinoma. The gradual morphologic transformation parallels with stepwise accumulation of genetic and epigenetic alterations.² Neutral evolution and short periods of genomic instability may lead to the concomitant occurrence of several molecular alterations and contribute to CRC polyclonal landscapes.³⁻⁵

The molecular landscape of CRC has prognostic relevance and affects the choice of therapeutic strategies, directing the rational deployment of targeted drugs directed against deregulated cellular processes to which CRC cells are dependent for their survival and proliferation.

A key factor in determining CRC cell proliferation is aberrant activation of the epidermal growth factor receptor (EGFR) signaling pathway.^{6,7} Activation of EGFR, elicited by epidermal growth factor, leads to sequential activation of intracellular signaling proteins, such as the Kirsten rat sarcoma viral oncogene homolog (KRAS), the B-Raf serine/threonine-protein kinase (BRAF), and the extracellular signal-regulated kinase 1 (ERK1), conveying proliferative signals through regulation of gene expression.⁸ Standard treatment of patients with metastatic CRC (mCRC) is mainly based on cytotoxic chemotherapy with ad hoc addition of molecular-targeted regimens.⁹ For example, anti-EGFR monoclonal antibodies, such as cetuximab and panitumumab, are administered as first- or second-line therapy in combination with chemotherapy. Even with the combination of these drugs, the median overall survival of patients with mCRC does not go beyond 30 months.^{10,11} Furthermore, EGFR targeted inhibition is effective only in a molecularly-defined subgroup of patients. CRC tumors carrying activating mutations in *KRAS* or *BRAF* genes are usually refractory to EGFR blockade,^{12,13} and anti-EGFR monoclonal antibodies are approved only for treatment of *RAS/BRAF* wild-type tumors.¹⁴⁻¹⁶ Notably, a sizable fraction of wild-type cases are intrinsically resistant to anti-EGFR treatments, and their resistance is often associated with alterations in genes (such as *NRAS*, *ERBB2*, *EGFR*, *FGFR1*, *PDGFRA*, *MAP2K1*, or *MET*) that lead to downstream or parallel signaling activation.¹⁷⁻¹⁹ Unfortunately, even in responding patients, acquired resistance eventually emerges within 3 to 12 months of initiating therapies.²⁰⁻²³ From a molecular standpoint, the unsuccessful outcome of anti-EGFR therapy is mainly related to the emergence of mutations in the *EGFR-RAS* pathway.^{20,24}

In addition to EGFR blockade, we previously demonstrated in preclinical models that amplification of *ERBB2* (encoding the human epidermal growth factor receptor 2 [HER2] tyrosine kinase receptor) is an effective therapeutic target in cetuximab-resistant tumors.²⁵ Based on these observations, *HER2* Amplification for Colo-rectal cancer Enhanced Stratification (HERACLES), a phase II trial aimed at testing trastuzumab and lapatinib in patients with *ERBB2* amplified CRC, was performed with very encouraging results.²⁶

In addition, molecular alterations leading to the constitutive activation of other receptor tyrosine kinases also play a pivotal role

in colorectal tumorigenesis and drive primary resistance to EGFR targeted monoclonal antibodies. For example, fusions involving *ALK*, *RET*, *ROS1*, and *NTRK* family genes occur in 0.2% to 1% of the cases²⁷ and represent valuable therapeutic targets for highly selected patients with mCRC.^{28,29}

As previously mentioned, the CRC mutational landscape has also prognostic value. Mismatch repair proficient CRCs comprise 85% of the total cases and often arise in the left colon.³⁰ Mismatch repair deficient (MMRd) cancers that carry defects in the DNA repair machinery, and preferentially arise in the right colon, account for the remaining 15% of cases. MMR deficiency causes insertions and deletions in regions of repetitive DNA sequences called microsatellites. For this reason, MMRd often leads to the onset of a phenotype called “microsatellite instability,”^{31,32} which, importantly, is associated with favorable prognosis.^{33,34} Accordingly, the fraction of MMRd cases decreases to 5% to 7% in the metastatic setting.

In addition to molecular analyses performed on tissue samples, profiling circulating tumor DNA (ctDNA) offers unprecedented opportunities for genotyping, tracking minimal residual disease, and monitoring the emergence of drug resistance in CRC and other tumor types.³⁵ In light of its predictive role in response to EGFR blockade, the analysis of *RAS* mutational status in ctDNA from patients with mCRC has been recommended by the European Society of Digestive Oncology and the European Society for Medical Oncology when tumor tissue is not readily available.³⁶ Liquid biopsies also allow tracking of clonal evolution during treatment, for example, by detecting acquired alterations before disease progression is clinically manifest.²³ As compared with analyses performed on tissue, detection of mutant tumor DNA in blood is more challenging and requires dedicated methodologies.³⁵

For the reasons discussed above, defining the complex genomic landscape of CRCs and identification of genetically distinct CRC subtypes involve deep molecular characterization of individual patients. This is necessary for diagnostic purposes and to properly tailor treatments. In this work, we describe multiple DNA next generation sequencing (NGS) approaches that, coupled with computational and bioinformatic algorithms, allow determination of clinically relevant parameters in this clinical setting.

Materials and Methods

Patient Samples

Patients with mCRC were treated with a dual HER2 blockade by trastuzumab and lapatinib within the HERACLES multicenter, open-label, phase II trial performed at 4 academic cancer centers in Italy, as previously described.^{26,37} The study was conducted according to the provisions of the Declaration of Helsinki and the International Conference on Harmonization and Good Clinical Practice guidelines. All patients provided written informed consent for participation to the study and associated procedures, including the molecular analyses described in this work.

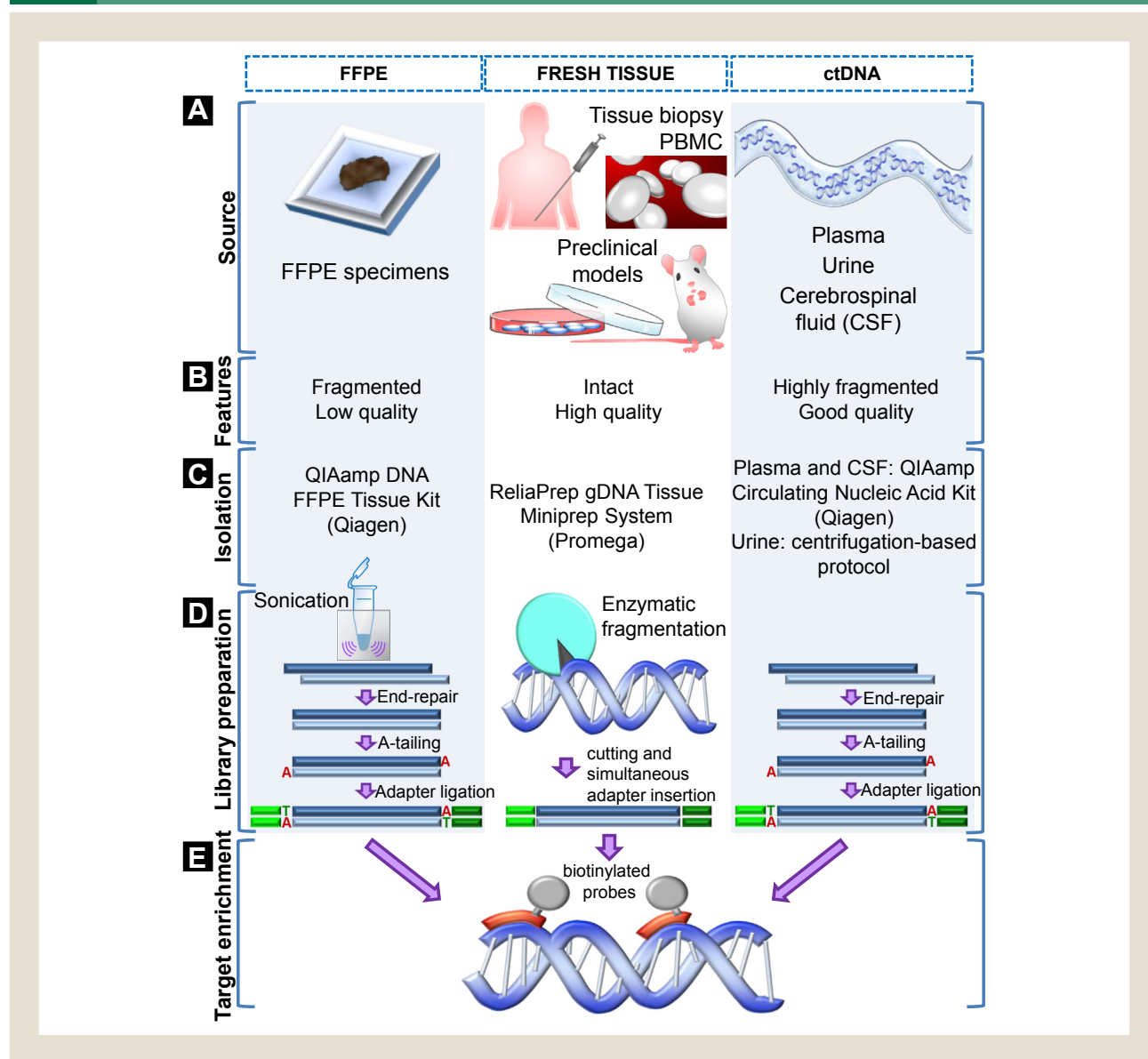
NGS: Target Enrichment and Custom Panel Design

We carried out and optimized specific workflows for both DNA extraction and the initial steps of library preparation (see [Supplemental Material](#) section in the online version) based on DNA-specific features ([Figure 1A, B](#)).

Independently from these initial steps, all libraries proceeded through enrichment of target regions. For whole exome sequencing (WES, 30 Mb), we used Nextera Rapid Capture Exome kit, or the latest equivalent TruSeq Rapid Exome Enrichment kit (Illumina Inc). For the IRCC-TARGET and FUSION custom panels, we designed capture probes exploiting the DesignStudio tool available online (<https://designstudio.illumina.com>). In particular, for the IRCC-TARGET panel, we identified a target covering all coding regions of 224 genes known to be involved in CRC tumorigenesis, progression, oncogenic signaling and sensitivity, or resistance to targeted therapy, for a total of 603 kb (see [Supplemental Table 1](#) in the online

version). Instead, the FUSION panel has been designed selecting the most frequent oncogenic kinases involved in fusions and the most frequently rearranged partners, identified on the basis of the available literature and The Cancer Genome Atlas database. Custom probes were designed to capture exons and introns of upstream (5') and downstream (3') partners. The panel also allowed enriching hot-spot mutations previously associated to EGFR blockade resistance in CRC, the entire promoter of EGFR ligands, and, finally, all coding exons of genes known to be involved in CRC tumorigenesis (*PTEN*, *TP53*, *APC*, *CTNNB1*). The entire selected target regions encompassed 918kb (see [Supplemental Table 2](#) in the online version). Upon

Figure 1 Customized Workflow for DNA Extraction and Next Generation Sequencing Library Preparation. A, Different Sample Types are Typically Available, Including FFPE, Fresh Tissue, or ctDNA. B, Based on the Tissue of Origin, Colorectal Cancer DNA Presents Different Features. C, Definite Protocols for DNA Extraction are Used on the Basis of DNA Characteristics. D, Library Prep Steps Have Been Optimized According to DNA Features, to Obtain DNA Fragments With Proper Length and to Include Sequencing Adapters. E, the DNA Enrichment Step is the Same Regardless of the Type of Sample: DNA Regions of Interest Hybridize With Target-specific Biotinylated Probes and are Then Captured by Streptavidin-Coated Beads



Abbreviations: CSF = cerebrospinal fluid; ctDNA = circulating tumor DNA; FFPE = formalin-fixed paraffin-embedded; PBMC = peripheral blood mononuclear cell.

quality assessment, final libraries were then sequenced using Illumina MiSeq or NextSeq500 sequencers (Illumina Inc).

Bioinformatic Analysis of NGS Data

All bioinformatic tools were run with default parameters unless otherwise specified. In the quality control (QC) module and “Mapping” phase, raw reads generated by the sequencer were aligned to the reference genome by *bwa-mem*³⁸ algorithm (version 0.7.13-r1126); polymerase chain reaction (PCR) duplicates were marked using MarkDuplicates in the Picard tools suite³⁹ (v. 2.0.1). SAMtools⁴⁰ (v. 1.3.1) was used for reading, writing, or viewing files in the SAM/BAM/CRAM format. The circular binary segmentation algorithm, as implemented in the DNA-copy R module,⁴¹ was used to cluster all gene copy-number alterations (CNA) in the dedicated module. Pindel⁴² (v. 0.2.5b6) was used for local read realignment in the insertion/deletion (INDEL) module. Blat⁴³ (v. 35) was used for fine remapping of the reads in the FUSION module, with *tileSize* = 11 and *stepSize* = 5. We set that each reported fusion breakpoint must be supported by at least 10 reads and each fusion partner must have at least 15 mapped bases on the respective end of the read.

To carry out analyses for multiple patients at the same time, the bioinformatic workflow leverages a high performance computing cluster composed of 5 nodes running the SLURM workload manager. The use of a high performance computing cluster allows spreading jobs across nodes to significantly speed-up analysis as well as storing in a central location sequencing data, genome references, aligner indexes, annotations, genomic databases, and analysis tools to ensure reproducibility. All custom scripts are available at <https://bitbucket.org/ircct/idea>.

Results

NGS of CRC Samples

The choice of sample types to be analyzed by NGS is dictated by availability or clinical questions. For instance, most often formalin-fixed paraffin-embedded (FFPE)-derived DNA samples are available for retrospective studies, whereas tumor heterogeneity, or longitudinal monitoring and detection of minimal residual disease, is often assessed using plasma ctDNA.^{23,44} Sometimes, fresh tissues (such as biopsies and preclinical models) can also be available (Figure 1A). It is therefore of utmost relevance to be able to process a variety of different samples. To this aim, we outline below specific guidelines for the initial steps of sample preparation.

First of all, we identified nucleic acid isolation as a crucial step that requires tailored protocols and specific kits for each sample type (Figure 1C) to generate DNA of suitable quality and quantity for further analyses (Figure 1B).

After DNA extraction, samples authentication with short tandem repeat analysis is performed to avoid misidentification and to verify the correct matching among samples belonging to the same patient/preclinical model.

Because DNA displays distinct characteristics depending on the starting material (Figure 1B), we adapted and optimized sample processing and sequencing protocols according to sample types.

In our experience, FFPE-derived DNA typically shows poor quality, likely associated with the processing steps of histology specimen preparation. In addition, the presence of DNA fragments of variable length makes this type of sample the most challenging to

process with the NGS workflow. We obtained a remarkable improvement in the quality of final sequencing results by mechanical shearing of DNA, thus rendering fragment length homogeneous and tailored for Illumina sequencers. The next steps involve end-repair and A-tailing of fragmented DNA molecules, both essential for subsequent ligation of adapter sequences (Figure 1D).

Unlike FFPE-derived DNA, ctDNA displays good quality owing to the absence of chemical contaminants, but it is highly fragmented. In light of this, in our protocol, ctDNA is directly subjected to end-repair and A-tailing steps prior to adapter ligation (Figure 1D).

Finally, intact high-quality DNA is usually isolated from fresh or frozen tissue. We enzymatically fragment this type of DNA by means of a transposon that cuts and simultaneously inserts sequencing adapters (Figure 1D).

In all cases, index sequences specific for each sample are inserted by means of a short PCR amplification, thus allowing to pool several samples in the same library.

The approaches outlined above allow isolation of DNA fragments of suitable length with ligated adapters. At this point, all samples could be directly subjected to whole genome sequencing (WGS). However, as compared with WGS, the analysis of specific regions of interest, such as WES or custom panels, provides several advantages, as discussed later. We found that the capture-based approach is the preferred choice for enrichment of target regions, because it introduces less intrinsic biases compared with amplicon-based strategies.⁴⁵ In capture-based approaches, specifically designed biotinylated probes that hybridize to the corresponding target sequences are then captured by streptavidin magnetic beads (Figure 1E). The enriched libraries are then subjected to a final short PCR amplification and, afterwards, loaded on the sequencer.

Genotyping CRCs in Blood

We and others have shown that liquid biopsies can complement and, in some instances, provide more information than standard tissue biopsies in patients with advanced CRC.³⁵ Analyses of plasma samples offer the possibility to obtain a broad range of information on tumor heterogeneity and clonal molecular dynamics from a blood withdrawal. Notably, the workload required for plasma processing takes significantly less time than preparation of FFPE samples (Figure 2A). Importantly, to preserve ctDNA in plasma, blood samples must be processed within 2 to 4 hours from collection. After blood centrifugation and plasma isolation, ctDNA can be extracted within a few hours. Overall, ctDNA can be available for downstream analyses within 24 to 36 hours from sample collection. This aspect is important as, in some instances, the timeline required to generate molecular maps starting from ctDNA or genomic DNA extracted from FFPE tissue has clinical relevance.

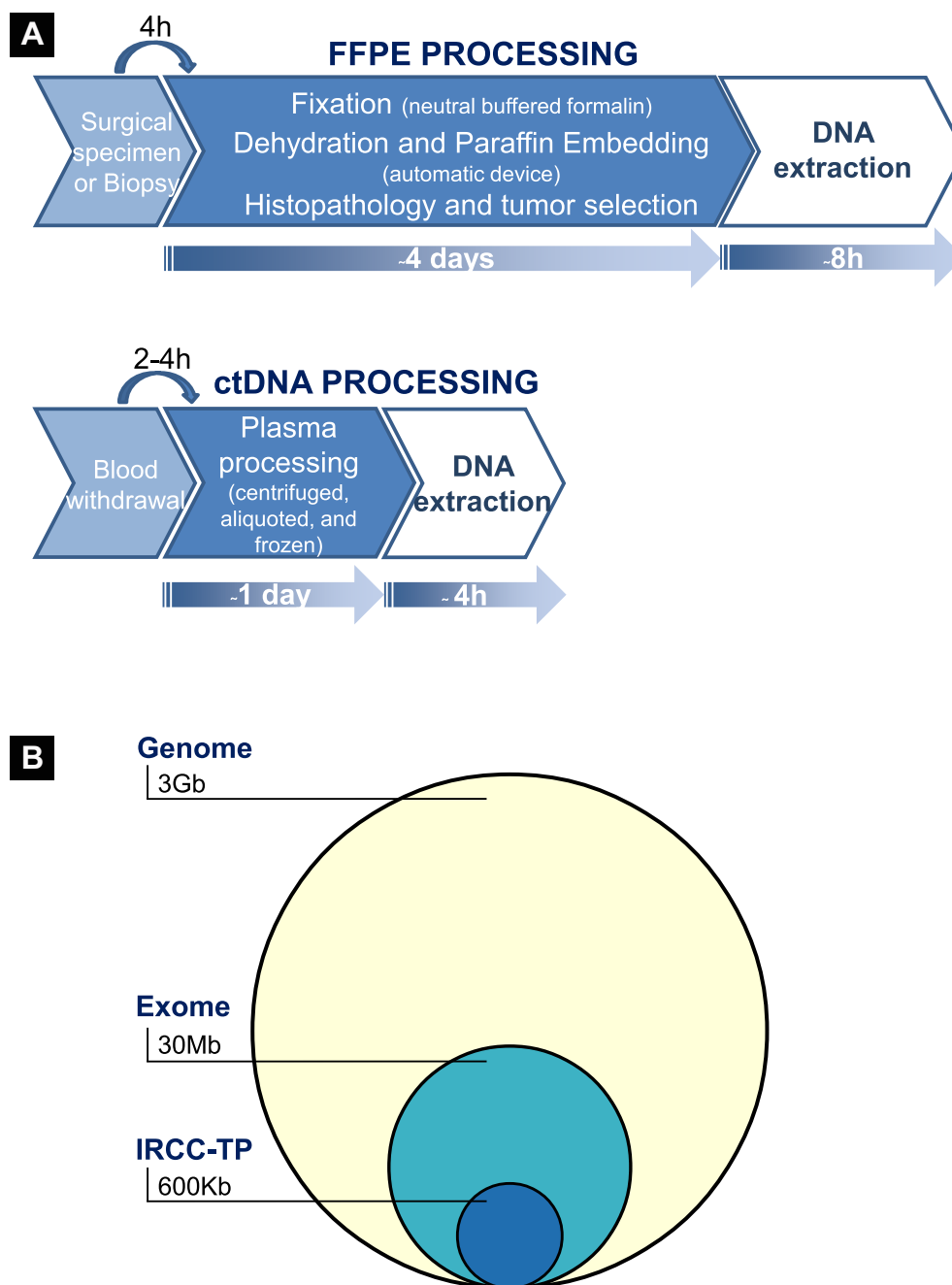
Target Choice for Optimal Sensitivity

Although the added value of NGS-based analysis in precision medicine is undisputed, optimal implementation of NGS strategies for diagnostic purposes is still being evaluated. The more commonly used NGS approaches are WGS, WES, and custom gene panels. Indeed, clinically actionable targets are typically localized in a small subset of genes,⁶ which renders sequencing of defined genomic regions a valuable and cheaper alternative to WES or WGS. However,

some features, such as determination of the microsatellite status (microsatellite stable/microsatellite instable) and of the tumor mutational burden, require sequencing of a sizeable fraction of the genome for reliable results, which may not always be achieved using targeted panels.⁴⁶

The most important technical difference between a custom gene panel and either WGS or WES is related to the minimum frequency of the mutant allele that can be reliably discriminated, thus defining the limit of detection. The smaller the target (Figure 2B), the greater the sequencing depth, thus allowing to better recognize low frequency

Figure 2 A, Time Required to Process Different Sample Types. The Picture Pinpoints the Processing Time for FFPE or Blood Samples. B, Comparison Between Different Targeted Region Sizes. Circles Represent the Relative Size of Each Target Defined in the Experimental Design. The Entire Human Genome is 3 Billion Base Long, Whereas the Coding Regions is 100 Times Smaller (About 30 Mb); IRCC-TARGET Panel Spans About 600kb and Contains Only Exonic Regions



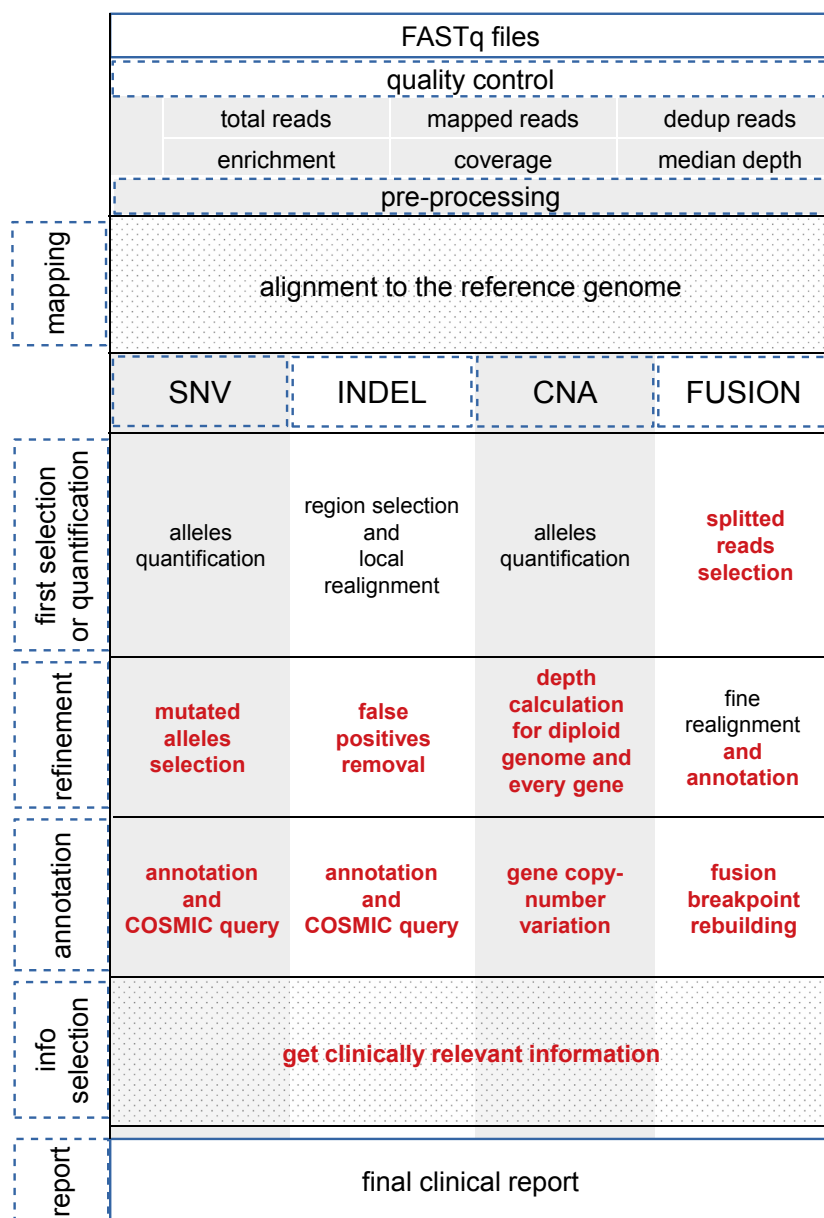
Abbreviations: ctDNA = circulating tumor DNA; FFPE = formalin-fixed paraffin-embedded.

IDEA NGS Workflow

variations from noise, that is, owing to errors introduced by sample preparation and base calling.⁴⁷ This is particularly relevant for liquid biopsy sample analysis: in plasma, tumor-derived DNA shed in the bloodstream is diluted by DNA released by normal tissue⁴⁸; accordingly, the optimal limit of detection for ctDNA should be below 1%. Other advantages of custom panels are related to the smaller number of sequenced bases, which allows for faster, cheaper, and less time-demanding analyses compared with WES or WGS.

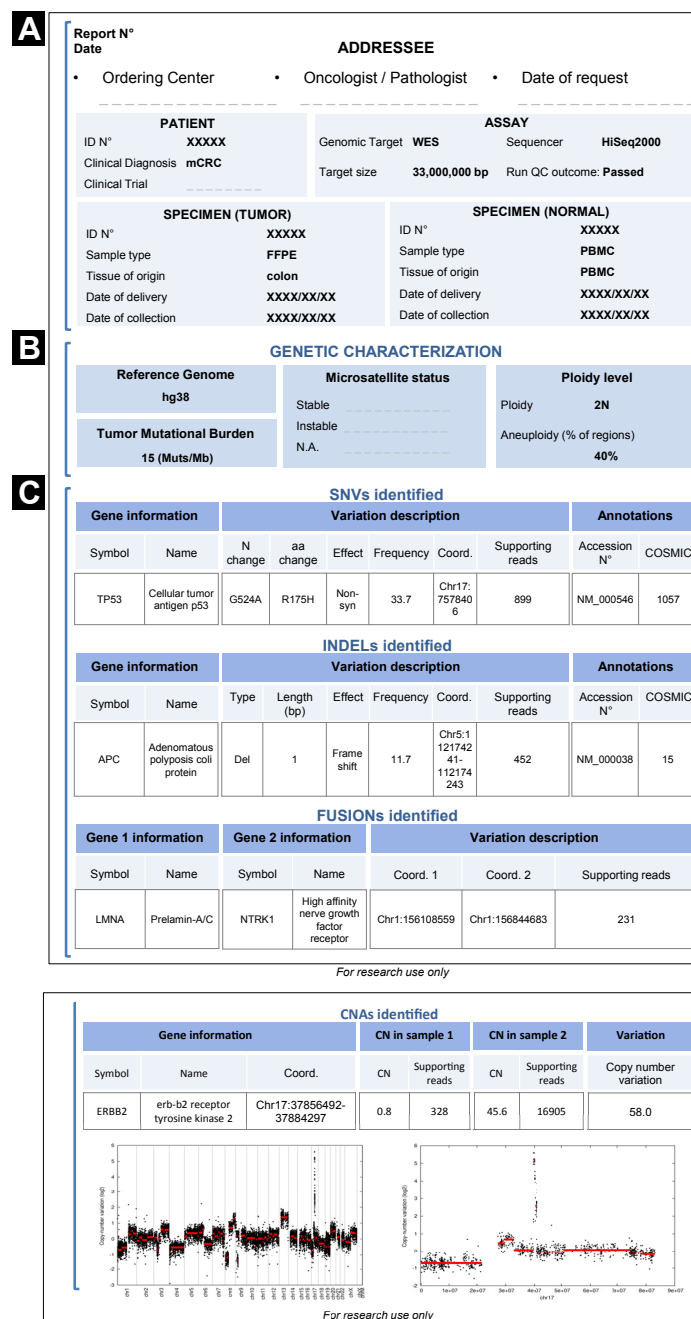
Over the past 5 years, we developed 2 target panels called IRCC-TARGET and FUSION panels. The IRCC-TARGET panel was designed to identify alterations in genes that are frequently mutated in CRC,²³ whereas the FUSION panel is focused on the identification of translocations in CRC samples (for full description of custom panel targets, see the Materials and Methods section). The FUSION panel is larger, owing to the need of sequencing intron-exon junctions to precisely identify the genomic breakpoint where

Figure 3 Bioinformatic Workflow From the Sequencing Output to the Final Report. The Sequencer Output Raw Data are Preprocessed (Quality Control Step) and the Parameters Shown in the Figure are Evaluated. Then the Data is Processed Through up to 4 Modules Depending on the Experimental Design. Initially, a Mapping Step is Performed, Then Each Module Exploits Either Public Tools (in Black) or Custom Scripts (Highlighted in Red). A Common Final Report is Generated, Encompassing Molecular and Clinically Relevant Information



Abbreviations: CNA = copy number alteration; COSMIC = Catalogue Of Somatic Mutations In Cancer; INDEL = insertion/deletion; SNV = single nucleotide variant.

Figure 4 The Final Report. A, The First Part of the Final Report Reviews General Information About the Patient, Along With Sample Characteristics and Next Generation Sequencing Technical Specifications. B, The Second Part Summarizes Overall Molecular Features of the Sample, Such as Tumor Mutational Burden, Microsatellite Status, and Ploidy. C, In the Third Part, Results From Each Module Used During the Bioinformatics Analysis are Reported in a Table Along With the Most Relevant Annotations. SNV and INDEL are Reported Listing Genomic Position, COSMIC Occurrence, Variant Frequency, Amino Acid, and Nucleotide Changes. FUSION are Reported Listing the 2 Genes Involved in the Translocation Together With the Number of Supporting Reads. CNAs are Shown Through a Graphical Representation of the Entire Genome and a More Detailed Plot of the Individual Chromosome



Abbreviations: CNAs = copy number alterations; COSMIC = Catalogue Of Somatic Mutations In Cancer; FFPE = formalin-fixed paraffin-embedded; INDEL = insertion/deletion; mCRC = metastatic colorectal cancer; PBMC = peripheral blood mononuclear cell; QC = quality control; SNV = single nucleotide variant; WES = whole exome sequencing.

the translocation occurs.⁴⁹⁻⁵¹ Once a translocation is identified, it can be exploited to design patient-specific PCR probes and track the rearrangement in ctDNA in longitudinal plasma samples.⁵¹

IDEA: Bioinformatic Workflow

The genomic landscape of CRCs encompasses several types of molecular alterations. We designed a comprehensive pipeline that is organized in specific modules (see [Supplemental Material](#) section in the online version), allowing for the identification of genomic alterations that are commonly associated with tumor onset, progression, and drug resistance such as: (1) single nucleotide variants (SNVs), (2) INDEL, (3) gene copy-number alterations (CNAs), and (4) fusions (FUSION). We assembled all the corresponding pipelines under a unique package, in which every module can be run independently, based on scientific or clinical requests ([Figure 3](#) and [Materials and Methods](#) section). Furthermore, every section of the pipeline can be separately deployed, modified, or upgraded.

The first step is the QC module, which is mandatory for any type of analysis, as it verifies whether the quality of the sequencing data is appropriate to move forward with the subsequent bioinformatic analyses. Depending on the type of starting material, the NGS assay used and clinical needs, QC parameters are defined case-by-case. The next step (which is optional) is the data pre-processing module, which is aimed at removing or trimming unrelated sequences. Mapping of raw reads to the reference genome and removal of PCR duplicates are then performed, representing a 2-step process common to all analyses.

After this, IDEA proceeds to the identification of tumor-specific molecular alterations. The overall aim is to detect somatic variations (ie, variations that are related to the disease and not to germline). For this reason, all these analyses are typically performed following a comparison strategy, in which variations found in the germline DNA (healthy tissue or peripheral blood mononuclear cells [PBMCs] from the same patient) are subtracted from those present in the matched tumor sample. The only exception to this strategy relates to the FUSION module, in which we assume that germline DNA does not carry rearrangements and a single sample analysis is performed. Importantly, in those cases in which the normal sample is not available, we routinely assess the single nucleotide polymorphisms database⁵² in order to filter out known germline alterations. Because they can have clinical relevance, germline variants of known pathogenic impact (such as *BRCA1*, *BRCA2*, *MLH1*) are also monitored and can be reported upon request.

The SNV module includes custom scripts to identify and annotate each mutation, also providing genomic information for the final report. Specifically, amino acid and nucleotide changes, along with isoform accession number, are reported. We also include the number of occurrences of the specific variant in the Catalogue Of Somatic Mutations In Cancer (COSMIC) database.⁵³

In the INDEL module, after comparing germline and tumor samples, the data are annotated, and INDELs present only in the tumor sample and whose allelic frequencies are predefined with a customizable threshold are listed. The COSMIC occurrence of individual INDELs is also reported.

The CNA module is designed to detect amplification or deletion of genomic regions in the tumor sample with respect to the matched germline. A custom algorithm clusters all the CNAs, allowing definition of contiguous regions with similar increase or loss of copy number.

The FUSION module has been devised to pinpoint chromosomal rearrangements and accurate genomic DNA breakpoint; furthermore, a specific algorithm selects only fusions that could be correctly translated.

The last step of IDEA generates a FINAL REPORT ([Figure 4](#)) which has been designed to provide the most clinically relevant information at a glance and is tailored for a clinical readership. Additional details are made available for in-depth review of complex cases. In the FINAL REPORT, clinical information is listed, as well as sample characteristics and the technical specifications (ie, NGS assay) ([Figure 4A](#)). The second part of the report summarizes overall genetic features of the analyzed sample, such as tumor mutational burden, microsatellite status, and ploidy ([Figure 4B](#)). Results are combined in tables based on the type of molecular alterations ([Figure 4C](#)). For each variant, we first indicate the gene carrying the alteration, following the HUGO Gene Nomenclature Committee-approved gene nomenclature (symbol and full name). This, together with the specific isoform accession number, avoids possible misunderstandings. Variants are sorted by their occurrence in the COSMIC database, highlighting previously identified and potentially relevant alterations. Specific features for each DNA variation are listed as well: for instance, in the mutational analysis, we report the SNV genomic position, the amino acid and nucleotide change, the substitution effect (synonymous, non-synonymous, or stop gain/loss), and the allele frequency. Finally, the read depth (supporting reads) is indicated to evaluate the accuracy of variation calls. In the INDEL module, the entire region affected by the variation is listed, together with its length, the effect, allele frequency, and read depth. The FUSION module reports the partner genes involved in the translocation event, the exact genomic breakpoint, and the read support. Finally, for the CNA module, numerical gene copy-number for both normal and tumor samples is indicated, and its fold-change in the tumor is also reported. A graphical representation of the whole genome CN segmentation has been implemented, to better visualize amplifications and losses at the gene and/or chromosome levels ([Figure 4](#)).

Deploying IDEA to Genotype Colorectal Tumors in Tissue and Blood

We previously reported the HERACLES clinical trial,²⁶ which was aimed at targeting HER2 with trastuzumab and lapatinib in *ERBB2*-amplified mCRC. Although 30% of the patients initially responded,²⁶ acquired resistance occurred in most of the cases. In collaboration with Guardant Health, we previously analyzed the blood of patients recruited in the HERACLES trial to identify putative mechanisms of primary and secondary resistance to trastuzumab and lapatinib.³⁷ To test the capabilities of the IDEA pipeline in a clinical setting, we performed NGS analyses on a subset of surgical tissue specimens and plasma samples collected during HERACLES treatment. In particular, we profiled using the IRCC-TARGET panel, FFPE samples, and ctDNA from 10 patients belonging to the HERACLES cohort. The bioinformatic pipeline was deployed to infer gene copy-number status of tissues from patients with CRC. Targeted capture sequencing allowed to reach high read depth (median value of 222×) and fraction of covered target (87.9% with at least 10 reads) in each sample, providing high-quality and reliable results. CN analysis revealed

high-level of *ERBB2* amplification in 8 of 10 patients (CNA greater or equal to 3) (Figure 5) and a more modest increase in the other 2 patients. The analysis did not reveal recurrent copy number alterations in other genes (Figure 5).

Next, we studied plasma samples to unveil possible variations associated with drug resistance. In patients who first achieved clinical benefit, but then relapsed, ctDNA profiles at progression and baseline were compared, whereas for progressive disease cases, only the baseline ctDNA samples were analyzed (Figure 6). Using the IRCC-TARGET panel, we were able to identify genetic alterations in the ctDNA of all patients, including trunk variations in *TP53* and *APC*. We also detected drug-resistance-related mutations in *KRAS*, *BRAF*, *PIK3CA*, and *ERBB2* in 5 of 10 patient samples (Figure 6). Alterations in *RAS/RAF* highlighted the importance of the MAPK pathway as key mediator of resistance to anti-HER2 therapies. Among patients who experienced clinical benefit, emerging *KRAS* mutant clones and *BRAF* amplification were identified at progression. Other alterations detected at progression involved *ERBB2*, *EGFR*, *PIK3CA*, and *PTEN* (Figure 6), suggesting an involvement of the PI3K-AKT pathway in the acquisition of resistance to dual HER2 blockade.

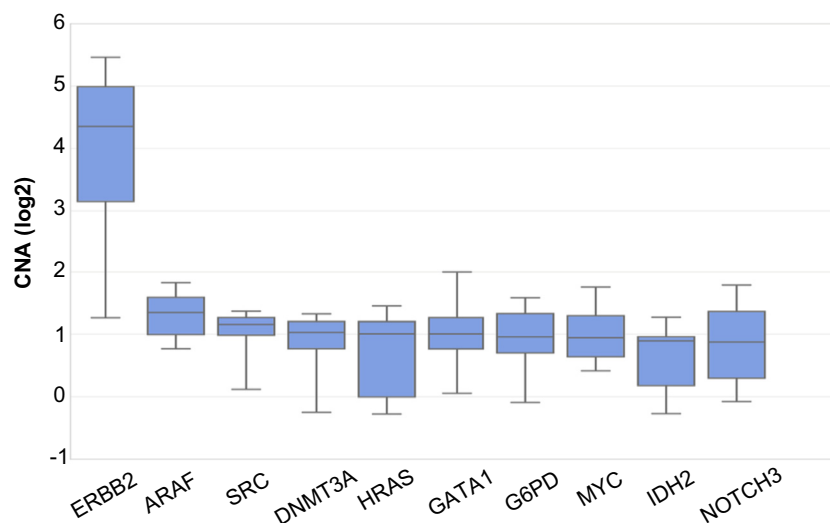
Discussion

It is now widely accepted that alterations in the DNA sequence underlie the development of neoplasms. The identification of mutated genes that are causally implicated in oncogenesis ('cancer genes') has been a major goal in medical sciences for the past 3 decades. The availability of the human genome sequence, coupled with the introduction of high throughput sequencing technologies, has created an unprecedented opportunity in the field of oncology.

It is now possible to generate accurate genomic profiles from DNA isolated from cancer tissues and from blood. NGS technologies are already available in cancer centers and academic institutions. Considering that sequencing costs have dropped significantly in the last 5 years and are projected to further reduce, NGS-based diagnostic assays are becoming widely applicable and have been entering clinical practice over the past few years. However, translating NGS data (raw sequencing 'reads') into a format that can be readily interpreted by pathologists and medical oncologists remains challenging and has not yet been standardized. To address this need, and using CRC as a test bed, we developed IDEA, a comprehensive analytical and computational pipeline. IDEA was conceived to identify somatic variants through the comparison of tumor and germline DNA samples. In this work, we have detailed each of the steps that, starting from a tissue fragment or a blood draw, are required to generate, process, and analyze NGS data.

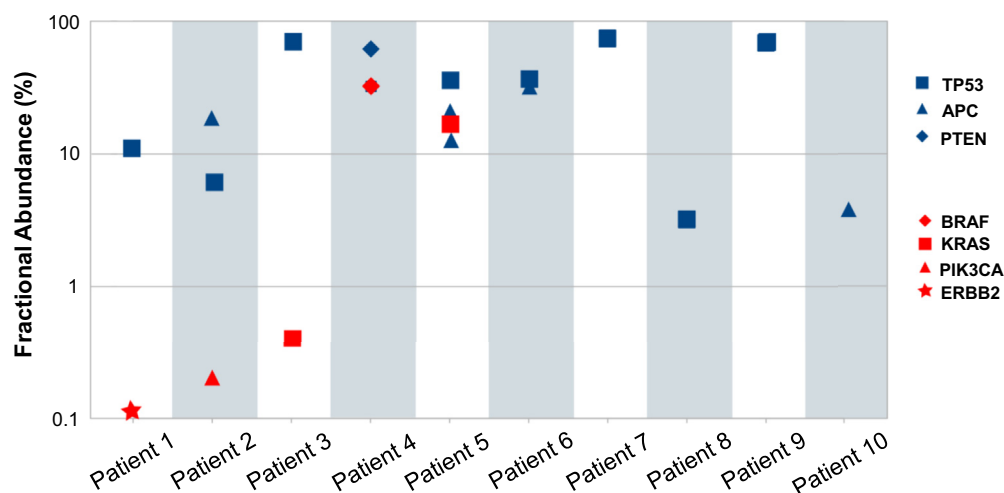
We show that IDEA can comprehensively identify several types of genetic alterations with clinical relevance including: single nucleotide variants, insertions and deletions, gene copy number alterations, and rearrangements. To test IDEA in the clinical setting, we exploited an annotated set of samples from patients with mCRC collected during the clinical trial HERACLES. In this phase II experimentation, patients positive for HER2 overexpression showed a response rate of 30% to trastuzumab and lapatinib,²⁶ a paradigmatic example of precision oncology. When the IDEA workflow was applied to DNA extracted from tissue and blood samples collected within HERACLES trial, it could readily pinpoint *APC* and *TP53* mutations as well as *ERBB2* copy number alterations. When patients progressed, IDEA could detect the emergence of *KRAS* and *BRAF* mutant clones as likely mechanisms of acquired

Figure 5 Copy-Number Landscape for *HER2* Amplification for Colo-Rectal Cancer Enhanced Stratification Patients as Defined by the IRCC-TARGET Panel. The Median of the Gene Copy-Number Distribution Across the Entire Dataset Was Used to Identify CNA in Individual Samples. For Plotting Purposes, Only the First 10 Genes are Shown



Abbreviation: CNA = copy number alteration.

Figure 6 Allele Frequencies Detected in ctDNA Samples of HERACLES Patients. Fractional Abundances of Driver and Drug-Resistance Genes in ctDNA of 10 HERACLES Patients are Shown. Colon Cancer Driver Genes are Shown in Blues (*APC*, *TP53*, and *PTEN*), Whereas Drug-Resistance Genes are Depicted in Red (*BRAF*, *KRAS*, *PIK3CA*, and *ERBB2*)



Abbreviation: ctDNA = circulating tumor DNA.

therapy resistance. In summary, we developed and clinically validated a start-to-finish analytical and computational pipeline to analyze tissue and blood samples from patients with CRC.

Conclusion

We have developed and integrated an experimental and bioinformatic pipeline (IDEA) to support diagnosis and clinical management of colorectal cancer patients. Using DNA NGS and bioinformatic approaches, IDEA determines SNVs, insertion/deletions, CNAs, and gene fusions, and ultimately provides a user-friendly report of clinical utility.

Clinical Practice Points

- Precision oncology is based on the concept that tumor-specific genomic landscapes provide important prognostic and predictive information. Therefore, there is an urgent need for developing NGS and bioinformatic workflows that can be routinely used in clinical settings. Furthermore, profiling ctDNA offers unprecedented opportunities for early detection, tumor genotyping, tracking minimal residual disease, and cancer evolution in CRC and other tumor types.
- We developed IDEA, which integrates DNA NGS approaches and computational/bioinformatic algorithms, to precisely identify clinically relevant information in tissue and liquid biopsy of patients with CRC. These include SNVs, INDELs, gene CNAs, and fusions. To test IDEA in the clinical setting, we exploited an annotated set of samples from patients with mCRC collected during the clinical trial HERACLES, unveiling primary and secondary mechanisms of resistance.

- IDEA is a start-to-finish wet and computational pipeline to analyze tissue and blood samples from patients with CRC. IDEA generates a user-friendly report with clinical utility.

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Disclosure

A. Bardelli attended Guardant Scientific Advisory Boards. The remaining authors have stated that they have no conflicts of interest.

Supplemental Data

Supplemental material and tables accompanying this article can be found in the online version at <https://doi.org/10.1016/j.clcc.2019.02.008>.

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Supplemental Material

DNA Extraction and Library Preparation for Next Generation Sequencing (NGS)

DNA has been extracted from fresh tissue and blood-isolated peripheral blood mononuclear cells (PBMCs) by ReliaPrep gDNA Tissue Kit (Promega) and from formalin-fixed paraffin-embedded (FFPE) samples by means of QIAamp DNA FFPE Tissue Kit (Qiagen). Circulating tumor DNA (ctDNA) has been extracted from plasma and *cerebrospinal fluid* using the QIAamp Circulating Nucleic Acid Kit (Qiagen) according to the manufacturer's instructions. For urinary DNA (trDNA) extraction, a customized protocol has been developed.¹ In detail, urine has been concentrated to 4 mL using Vivacell 100 concentrators (Sartorius Corp) and incubated with 700 µL of Q-sepharose Fast Flow quaternary ammonium resin (GE Healthcare). Tubes were spun to collect sepharose and bound DNA. The pellet has been re-suspended in a buffer containing guanidinium hydrochloride and isopropanol, and the eluted DNA collected as a flow-through using polypropylene chromatography columns (Bio-Rad). The DNA was further purified using QIAquick columns (Qiagen). Fragment size distribution for both ctDNA and trDNA was assessed using the 2100 Bioanalyzer High-Sensitivity DNA assay kit (Agilent Technologies) according to the manufacturer's instructions, while the quantification was performed by means of Qubit dsDNA HS Assay Kit (ThermoFisher Scientific).

Before proceeding with the NGS workflow, authentication of each sample has been performed using PowerPlex 16 HS System (Promega), which interrogates short tandem repeats at 16 different loci (D5S818, D13S317, D7S820, D16S539, D21S11, vWA, TH01, TPOX, CSF1PO, D18S51, D3S1358, D8S1179, FGA, Penta D, Penta E, and amelogenin). Amplicons from multiplex polymerase chain reactions (PCRs) were separated by capillary electrophoresis (3730 DNA Analyzer, Applied Biosystems) and analyzed using GeneMapper v.3.7 software (Life Technologies).

Library preparation was performed starting from up to 300 ng of FFPE-derived DNA, 150 ng of ctDNA, and 100 ng of gDNA from fresh tissues or PBMC. Degraded and bad-quality FFPE-derived DNA was sheared with focused-ultrasonicator M-220 (Covaris), with specific setting for 300 bp peak fragment distribution (peak incident power 50W, Duty factor 20%, cycles per burst 200, time 65 seconds, temperature 20°C). Afterwards, FFPE-derived DNA workflow proceeded with the TruSeq Nano DNA Library Prep kit (Illumina Inc.) for End-repair, A-tailing, and adapter ligation steps. For ctDNA, these steps were performed by means of NEBNext Ultra DNA Library Prep Kit for Illumina (New England BioLabs Inc.). gDNA from fresh tissue or PBMCs was fragmented using transposons, adding simultaneously adaptor sequences, thanks to Nextera Rapid Capture kits from Illumina.

Bioinformatic Modules for Analysis of DNA NGS Data

The quality control (QC) module defines a number of parameters that characterize the sequencing output (FASTQ files), such as the total number of raw reads and their percentage associated to each sample. Another critical output provided by the QC module is the number of reads that correctly map to the reference genome; this is then used to evaluate the amount of sequences suitable for further analyses. It is also important to evaluate the percentage of reads classified as PCR duplicates (identical copies) that should be removed because they introduce a bias in the analysis. Enrichment, coverage, and median depth are also evaluated with respect to the target regions of interest. In detail, the enrichment is the percentage of reads falling onto the target regions (on-target); depth represents the median number of reads covering a single genomic position; coverage represents the percentage of target actually captured and sequenced at a given depth.

The pre-processing module allows removing needless sequences. For instance, short sequences can be added to the DNA fragments during sample preparation based on the chemistry applied. In other instances, when DNA from human tumors implanted in immunocompromised mice (patient-derived xenografts) is analyzed, Xenome² (v. 1.0.1) is used to remove unwanted mouse-derived reads.

The single nucleotide variant (SNV) module is composed by a custom script that calculates the allele count by analyzing the alignment file. The algorithm then selects genomic coordinates carrying at least 1 SNV compared to the reference base. At this point, positions with low depth are discarded, and only variants with a predefined read support (at least 10, customizable) are called, thus minimizing sequencing errors.

The insertion/deletion (INDEL) module is based on the Pindel algorithm,³ which performs the local realignment of all reads and selects regions to search for insertions or deletions. Results are filtered out for hits that are recurrent across a wide number of samples and that most likely represent sequencing artifacts.

In the CNA module, the median read depth of the entire target is considered as the diploid level; the comparison with the median gene read depth allows estimating the gene copy number. CNA is calculated as the ratio between germline and tumor sample gene copy number.

The FUSION module extracts all reads that are already included in the mapping file and that show a non-perfect alignment, thus potentially harboring translocations. These selected reads are then mapped again using a more specific aligner. The output data are processed to detect translocated reads associated with fusions in which both (rearranged) partners can be correctly translated.

Supplemental Table 1 List of Genes Analyzed by the Custom IRCC-TARGET Panel

IRCC-TARGET Panel								
ABL1	CCND1	DNMT3A	FBXW7	HSP90B1	MEN1	NOTCH4	RAD51C	SRC
AIP	CCNE1	DPYD	FGFR1	IDH1	MET	NPM1	RAD51D	STAG2
AKT1	CDC73	EGFR	FGFR2	IDH2	MGMT	NRAS	RAF1	STK11
AKT2	CDH1	EPCAM	FGFR3	IGF1R	MLH1	NSD1	RB1	SUFU
AKT3	CDK4	ERBB2	FGFR4	IGF2	MLL	NTRK1	RECQL4	TCF7L1
ALK	CDK6	ERBB3	FH	IGF2R	MPL	NTRK2	RET	TCF7L2
AMER1	CDK8	ERBB4	FHIT	IKZF1	MRE11A	NTRK3	RHBDF2	TET2
APC	CDKN1C	ERCC1	FLCN	JAK1	MSH2	PALB2	RNF43	TGFBR2
ARAF	CDKN2A	ERCC2	FLT3	JAK2	MSH6	PAX5	ROS1	TMEM127
AR	CDKN2B	ERCC3	FOXL2	JAK3	MTHFR	PBRM1	RUNX1	TNFAIP3
ARID1A	CDKN2C	ERCC4	G6PD	KDR	MTOR	PDGFRA	SBDS	TP53
ARID1B	CEBPA	ERCC5	GALNT12	KIAA1804	MUTYH	PDGFRB	SDHAF2	TPMT
ASXL1	CEP57	EXT1	GATA1	KIT	MYC	PHOX2B	SDHB	TSC1
ATM	CHEK2	EXT2	GATA2	KRAS	MYCL1	PIK3CA	SDHC	TSC2
ATRX	CREBBP	EZH2	GNA11	LMO1	MYCN	PIK3CD	SDHD	TSHR
AURKA	CTNNB1	FANCA	GNAQ	MAP2K1	MYD88	PIK3R1	SEMA3D	TYMS
BAP1	CYLD	FANCB	GNAS	MAP2K2	NAV2	PMS1	SF3B1	UGT1A1
BLM	CYP1A2	FANCC	GPC3	MAP2K4	NBN	PMS2	SKP2	VHL
BMPR1A	CYP2C19	FANCD2	GREM1	MAPK11	NCOA3	POLE	SLX4	VKORC1
BRAF	CYP2C9	FANCE	HMGA1	MAPK12	NF1	POLD1	SMAD2	WRN
BRCA1	CYP2D6	FANCF	HMGA2	MAML1	NF2	PRF1	SMAD3	WT1
BRCA2	DAXX	FANCG	HNF1A	MAX	NKX2-1	PRKAR1A	SMAD4	XPA
BRIP1	DDB2	FANCI	HRAS	MDM2	NOTCH1	PTCH1	SMARCB1	XPC
BUB1B	DICER1	FANCL	HSP90AA1	MDM4	NOTCH2	PTEN	SMO	XRCC1
CBL	DIS3L2	FANCM	HSP90AB1	MED12	NOTCH3	PTPN11	SOX9	

Supplemental Table 2 List of Regions Targeted by the Custom FUSION Panel

FUSION Panel							
PARTNER 5'	KINASE 3'	PARTNER 5'	KINASE 3'	PARTNER 5'	KINASE 3'	KINASE 5'	PARTNER 3'
EML4	ALK	CCDC6	RET	TP53	NTRK1	FGFR3	TACC3
TFG	ALK	AKAP13	RET	TPM3	NTRK1	FGFR3	ELAVL3
TPM1	ALK	FKBP15	RET	MPRIIP	NTRK1	FGFR3	BAIAP2L1
TPM3	ALK	SPECC1L	RET	CD74	NTRK1	FGFR3	AES
TPM4	ALK	TBL1XR1	RET	NFASC	NTRK1		
STRN	ALK	ERC1	RET	BCAN	NTRK1		
SMEK2	ALK	KIF5B	RET	LMNA	NTRK1		
GTF2IRD1	ALK	NCOA4	RET				
KIF5B	ALK	PRKAR1A	RET				
KLC1	ALK	GOLGA5	RET				
VCL	ALK	ERC1	RET				
CEP85L	ROSI	TRIM24	RET				
SLC34A2	ROSI	TRIM27	RET				
CD74	ROSI	TRIM33	RET				
TPM3	ROSI	KTN1	RET				
SDC4	ROSI	PCM1	RET				
EZR	ROSI	HOOK3	RET				
LRIG3	ROSI	EIF3E	RSP02				
CCDC6	ROSI	PTPRK	RSP03				
GOPC	ROSI						
POINT MUTATIONS		ALL EXONS			5' UTR		
KRAS		PTEN			TGFA		
NRAS		TP53			AREG		
BRAF		APC			EREG		
PIK3CA		CTNNB1			EGF		
MEK							
EGFR							

Supplemental References

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